



BAULKHAM HILLS HIGH SCHOOL

2012
YEAR 11 YEARLY

Mathematics

General Instructions

- Reading time – 5 minutes
- Working time – 2 hours
- Write using black or blue pen
- Board-approved calculators may be used
- All necessary working should be shown in every question
- Marks may be deducted for careless or badly arranged work
- Attempt all questions

Total marks – 70

This paper consists of TWO sections.

Section 1 – Multiple Choice 10 marks

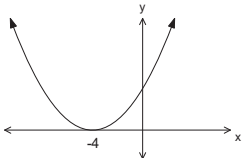
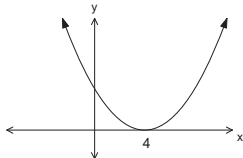
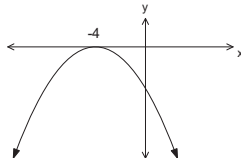
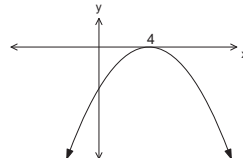
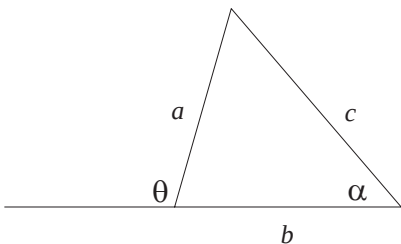
Section 2 – Extended Response 60 marks

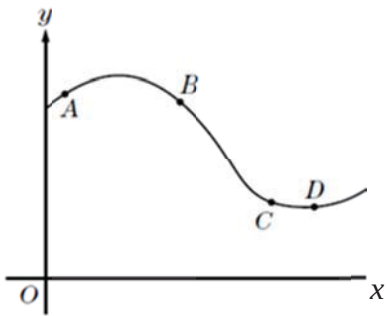
Attempt all questions

Start a new page for each question

Section 1 –Multiple Choice (10 marks)

Attempt all questions.

Answer the following on the answer sheet provided.		Marks
1	What is the solution of $ 1 - 2x < 7$? (A) $x < -3, x > 4$ (B) $x < -4, x > 3$ (C) $-3 < x < 4$ (D) $-4 < x < 3$	1
2	Which of the following could be the graph of $y = (4 - x)^2$? (A)  (B)  (C)  (D) 	1
3	Factorise fully: $x^2 - 4y^2 + 16y - 16$ (A) $(x + 2y - 4)(x - 2y - 4)$ (B) $(x - 2y - 4)(x + 2y + 4)$ (C) $(x - 2y + 4)^2$ (D) $(x - 2y + 4)(x + 2y - 4)$	1
4	If $a^b = 3$, then $a^{4b} - 5$ equals (A) 76 (B) 7 (C) 22 (D) 86	1
5	(2,5) is the midpoint of (5, y) and (x, 7). $x + y =$ (A) 2 (B) 3 (C) 4 (D) 18	1
6	If the equation $(1 - 3k)x^2 + 3x - 4 = 0$ has real roots, and k is an integer, what is the largest possible value for k ? (A) -2 (B) -1 (C) 0 (D) 1	1
7	If $a > b$, which statement is always true? (A) $a^2 > b^2$ (B) $\frac{1}{a} > \frac{1}{b}$ (C) $-a > -b$ (D) $2^a > 2^b$	1
8	Which of the following statements is true for this diagram?  (A) $c^2 = a^2 + b^2 + 2ab \cos \theta$ (B) $c^2 = a^2 + b^2 - 2ab \cos \theta$ (C) $a^2 = b^2 + c^2 - 2bc \cos \theta$ (D) $\frac{a}{\sin \alpha} = \frac{c}{\cos \theta}$	1

9	 At which of the points shown are y' and y'' both negative? (A) A (B) B (C) C (D) D	1
10	The area of a rectangle is 80 m^2 and its length is x metres. Its perimeter, in metres, is: (A) $2x + 160$ (B) $2x + \frac{80}{x}$ (C) $2x + \frac{40}{x}$ (D) $2x + \frac{160}{x}$	1
End of Section I		

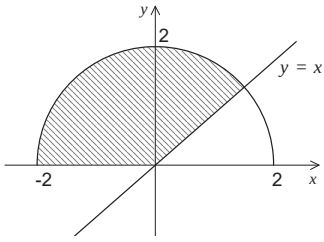
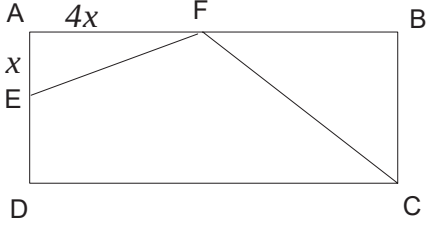
Section II – Extended Response

Attempt all questions. Show all necessary working.

Start each question on a new page. Clearly indicate question number.

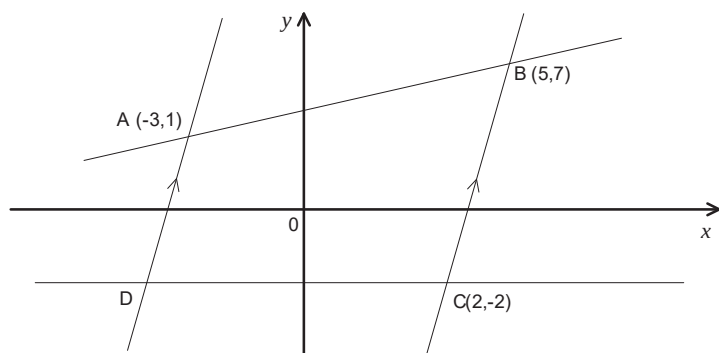
Write your name and teacher's name at the top of each new page.

Question 11 (15 marks) - Start a new page		Marks
a)	If $(2 + \sqrt{3})^2 = a + \sqrt{b}$, where a and b are rational, find the values of a and b	2
b)	Solve $8^{2-x} = 4^{2x}$	2
c)	Solve $\frac{3x-1}{4} - \frac{2x-5}{8} = x$	2
d)	Differentiate (i) $y = \frac{2x-1}{x+1}$ (ii) $f(x) = (2x-1)(3x+2)^5$	2 3
e)	State the domain and range of the function $f(x) = 2\sqrt{x+2} - 3$	2
f)	Evaluate $\lim_{x \rightarrow 2} \frac{4-x^2}{x^3-8}$	2

Question 12 (15 marks) - Start a new page		Marks
a)	Write the set of inequalities whose intersection describes the shaded region 	3
b)	Find the exact value of $\cos 240^\circ$	1
c)	Solve for $0^\circ \leq x \leq 360^\circ$: (i) $\cos 2x = \frac{1}{\sqrt{2}}$ (ii) $3\sin^2 x + 2\sin x - 1 = 0$	2 3
d)	ABCD is a rectangle with AB=12 cm and AD=6 cm. F is a point on AB such that AE=x and AF=4x. 	2 1
e)	If the function $f(x) = (k - 1)x^2 + (k + 2)x + 4$ is positive definite, find all possible values of k	3

Question 13 (15 marks) - Start a new page		Marks
a)	Find the equation of the tangent to the curve $y = x^3 - 3x$ at the point on the curve where $x = 2$	3
b)	Let α and β be the roots of the equation $2x^2 - 5x - 2 = 0$. Evaluate: (i) $\alpha + \beta$ (ii) $\alpha\beta$ (iii) $(\alpha - 2)(\beta - 2)$	1 1 1

- c) A, B and C are the points $(-3,1)$, $(5,7)$ and $(2,-2)$ respectively.
The equation of the line AB is $3x - 4y + 7 = 0$ and the equation of DC is $y = -2$.
AD is parallel to BC.

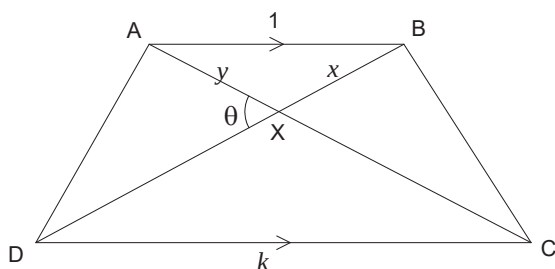


- (i) Find the length of AB
- (ii) Calculate the perpendicular distance from C to AB
- (iii) Find the gradient of BC
- (iv) Show that the equation of AD is $y = 3x + 10$
- (v) State the coordinates of D
- (vi) Find the area of quadrilateral ABCD

1
2
1
2
1
2

Question 14 (15 marks) - Start a new page

- a) For the curve $y = x^3 - 2x^2 - 4x$:
- (i) Find the stationary points and determine their nature
 - (ii) Show that there is a point of inflexion and determine its coordinates
 - (iii) Neatly sketch the curve clearly showing its important features (x -intercepts are not required)
- b) ABCD is a trapezium with $AB \parallel DC$, $AB=1$, $DC=k$ and $k > 1$. Its diagonals intersect at X, with $BX=x$ and $AX=y$.
Let $\angle AXD = \theta$.



- (i) Prove that $\triangle ABX$ is similar to $\triangle DCX$
- (ii) Explain why $DX = kx$ and $CX = ky$.
- (iii) Show that $AD^2 - BC^2 = (k^2 - 1)(x^2 - y^2)$
- (iv) Find the ratio $\frac{AD^2 - BC^2}{DB^2 - AC^2}$ in its simplest form

2
1
3
2

End of Examination

SOLUTIONS : Year 11 Yearly Exam 2012 (2-unit)

SECTION I

1. $-7 < 1 - 2x < 7$

$-8 < -2x < 6$

$4 > x > -3$

$-3 < x < 4$

(C)

2. (B)

3. $x^2 - (4y^2 - 16y + 16)$

$= x^2 - (2y - 4)^2$

$= (x - (2y - 4))(x + (2y - 4))$

$= (x - 2y + 4)(x + 2y - 4)$ (D)

4.

$(ab)^4 - 5 = 3^4 - 5$
 $= 76$ (A)

5. $\frac{5+x}{2} = 2$ $x = -1$

$\frac{y+7}{2} = 5$ $y = 3$

$\therefore x + y = -1 + 3 = 2$

(A)

6. $\Delta = 3^2 - 4(1 - 3k) \cdot -4$

$= 9 + 16 - 48k$

$= 25 - 48k \geq 0$

when $-48k \geq -25$

$k \leq \frac{25}{48}$

Largest integer $k = 0$ (C)

7. (D)

8. $c^2 = a^2 + b^2 - 2ab \cos(180^\circ - \theta)$

$= a^2 + b^2 - 2ab \cdot -\cos \theta$

$= a^2 + b^2 + 2ab \cos \theta$ (A)

9. Decreasing + conc. down (B)

10. $xy = 80 \therefore y = \frac{80}{x}$

$P = 2x + 2y = 2x + \frac{160}{x}$ (D)

SECTION II

Q11.

a) $(2 + \sqrt{3})^2 = 4 + 4\sqrt{3} + 3$

$= 7 + 4\sqrt{3}$

$= 7 + \sqrt{48}$

$\therefore a = 7, b = 48$

b) $(2^3)^{2-x} = (2^2)^{2x}$

$2^{6-3x} = 2^{4x}$

$6 - 3x = 4x$

$7x = 6$

$x = \frac{6}{7}$

c) $\frac{2(3x-1) - 1(2x-5)}{8} = x$

$6x - 2 - 2x + 5 = 8x$

$4x + 3 = 8x$

$3 = 4x$

$x = \frac{3}{4}$

d) (i) $u = 2x - 1$ $u' = 2$

$v = x + 1$ $v' = 1$

$\frac{dy}{dx} = \frac{vu' - uv'}{v^2}$

$= \frac{(x+1) \cdot 2 - (2x-1) \cdot 1}{(x+1)^2}$

$= \frac{2x+2 - 2x+1}{(x+1)^2}$

$= \frac{3}{(x+1)^2}$

(ii) $u = 2x - 1$ $u' = 2$

$v = (3x+2)^5$ $v' = 5(3x+2)^4 \cdot 3$
 $= 15(3x+2)^4$

$f'(x) = uv' + vu'$

$= \frac{15(2x-1)(3x+2)^4}{1} + \frac{2(3x+2)^5}{1}$

e) Domain: $x \geq -2$ ←
 Range: $f(x) \geq -3$ (or $y \geq -3$) ←

f) $\lim_{x \rightarrow 2} \frac{(2-x)(2+x)}{(x-2)(x^2+2x+4)} \leftarrow$
 $= \lim_{x \rightarrow 2} \frac{-(2+x)}{x^2+2x+4} \leftarrow$
 $= \frac{-4}{4+4+4}$
 $= -\frac{1}{3} \leftarrow$

Q12. $\left. \begin{array}{l} y \leq \sqrt{4-x^2} \\ \text{or } x^2+y^2 \leq 4 \end{array} \right\} \leftarrow$
 $y \geq x \leftarrow$

$y \geq 0$ (or $y > 0$) ←

b) $\cos 240^\circ = -\cos 60^\circ = -\frac{1}{2}$ ←

c) i) $2x = 45^\circ, 315^\circ, 405^\circ, 675^\circ$
 $x = 22.5^\circ, 157.5^\circ, 202.5^\circ, 337.5^\circ$

ii) $(3 \sin x - 1)(\sin x + 1) = 0$
 $\sin x = \frac{1}{3}$ or $\sin x = -1$

$x = 19^\circ 28', 160^\circ 32', 270^\circ$ ←

d) i) Area shaded

$= (12 \times 6) - \frac{1}{2} \cdot 4x \cdot x -$

$\frac{1}{2} \cdot (12-4x) \cdot 6$

$= 72 - 2x^2 - 3(12-4x)$

$= 72 - 2x^2 - 36 + 12x$ ←

$= \underline{36 + 12x - 2x^2}$

ii) Max. value of a quadratic fn occurs when $x = -\frac{b}{2a}$

ie. $x = \frac{-12}{2x-2} = 3$ ←
 [must have clear working]

e) For pos. definite $a > 0$, $\Delta < 0$

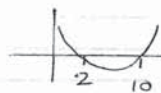
$\Delta = (k+2)^2 - 4(k-1) \cdot 4$

$= k^2 + 4k + 4 - 16k + 16$

$= k^2 - 12k + 20 \quad \text{--- (1)}$

$= (k-2)(k-10)$

0 when



$2 < k < 10$

but also $a = k-1 > 0$

$k > 1 \quad \text{(1)}$

$\therefore$ Solution $2 < k < 10$ (1)

Q13.

a) $y = x^3 - 3x$

$\frac{dy}{dx} = 3x^2 - 3$

$= 3(2)^2 - 3$ when $x = 2$

$= 9$

--- (1)

Point of contact is $(2, 2)$ ← (1)

$y - 2 = 9(x - 2)$

$y - 2 = 9x - 18$

$y = 9x - 16 \quad \text{--- (1)}$

Q13 (cont)

b) i) $\alpha + \beta = -\frac{b}{a} = \frac{5}{2}$

ii) $\alpha\beta = \frac{c}{a} = \frac{-2}{2} = -1$

iii) $(\alpha-2)(\beta-2)$

$= \alpha\beta - 2\alpha - 2\beta + 4$

$= \alpha\beta - 2(\alpha + \beta) + 4$

$= -1 - 2\left(\frac{5}{2}\right) + 4$

$= -2$

c) i) $AB = \sqrt{(5+3)^2 + (-7-1)^2}$

$= \sqrt{64 + 36}$

$= \sqrt{100}$

$= 10 \text{ units}$

ii) $d = \left| \frac{3(2) - 4(-2) + 7}{\sqrt{3^2 + (-4)^2}} \right|$

$= \left| \frac{21}{5} \right|$

$= \frac{21}{5} \text{ units.}$

iii) $m_{BC} = \frac{7+2}{5-2} = \frac{9}{3} = 3$

iv) $\therefore AD$ has $m=3$, and passes through $(-3, 1)$

$y - 1 = 3(x + 3)$

$y = 3x + 10$

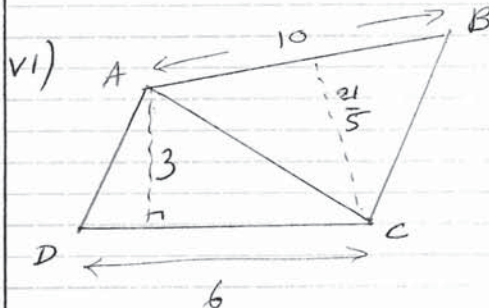
(1) v) At D: $y = -2$

$-2 = 3x + 10$

$3x = -12$

$x = -4$

$\therefore D(-4, -2)$



$A_1 = \frac{1}{2} \times 6 \times 3 = 9$

$A_2 = \frac{1}{2} \times 10 \times \frac{21}{5} = 21$

Total area = 30 units^2

[Also accept 36 units^2 , because given equation for AB is incorrect]

Q14.

a) $y = x^3 - 2x^2 - 4x$

i) $y' = 3x^2 - 4x - 4 = 0$ for stationary pts.

$$(3x+2)(x-2) = 0$$

$$\left. \begin{array}{l} x = -\frac{2}{3} \\ y = \frac{-40}{27} \end{array} \right\} \quad \left. \begin{array}{l} x = 2 \\ y = -8 \end{array} \right\}$$

$(-\frac{2}{3}, -\frac{40}{27})$ and $(2, -8)$ are stationary pts (1)

$$y'' = 6x - 4$$

At $(-\frac{2}{3}, -\frac{40}{27})$ $y'' = 6(-\frac{2}{3}) - 4 = -4 - 4 < 0$

$\therefore$ Maximum turning pt (1)

At $(2, -8)$ $y'' = 6(2) - 4 = 12 - 4 > 0$

$\therefore$ Minimum turning pt. (1)

ii) For a P.O.I., $y'' = 0$ and changes sign

$$y'' = 6x - 4 = 0 \quad \text{when} \quad x = \frac{2}{3}, \quad y = -\frac{88}{27}$$

$\therefore (\frac{2}{3}, -\frac{88}{27})$ is a possible P.O.I. $\leftarrow$ (1)

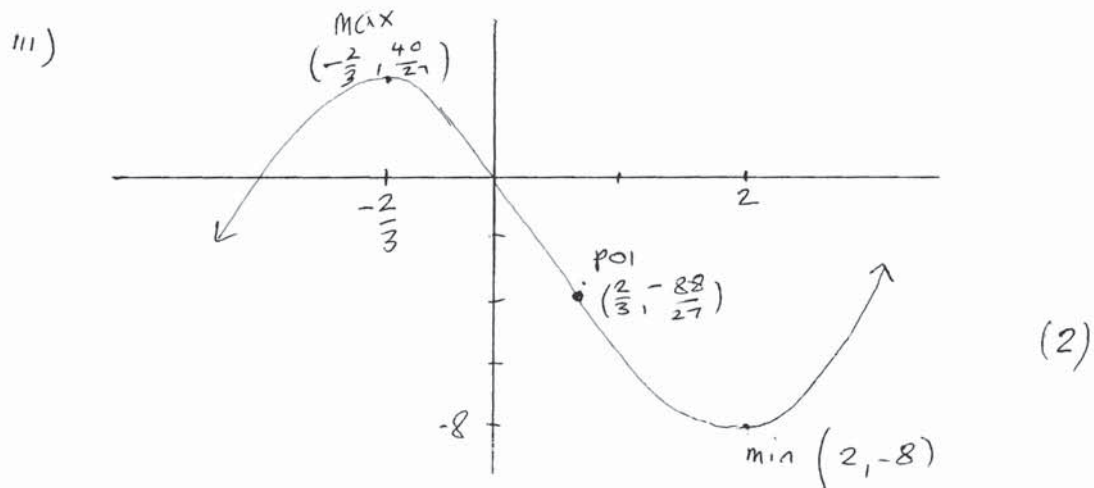
Now check

x	0	$\frac{2}{3}$	1
y''	-4	0	+2
	$\uparrow$		$\uparrow$
	$6(0) - 4$		$6(1) - 4$

Sign/concavity change

$\therefore (\frac{2}{3}, -\frac{88}{27})$ is a P.O.I.

F(1)



~~Question 15.~~

b) i) In $\triangle ABX$, $\triangle DCX$

$\angle AXB = \angle DXC$ (vertically opposite $\angle$ s equal)

$\angle BAX = \angle XCD$ (alternate $\angle$ s, $AB \parallel DC$) $\leftarrow (1)$

$\triangle ABX \parallel \triangle DCX$ (matching $\angle$ s equal) $\leftarrow (1)$

ii) $\frac{DX}{x} = \frac{CX}{y} = \frac{k}{1}$ (matching sides of similar $\triangle$ s proportional) $\leftarrow (1)$

$\therefore DX = kx$ and $CX = ky$

iii) $AD^2 = (kx)^2 + y^2 - 2 \cdot kx \cdot y \cdot \cos \theta$

$= k^2 x^2 + y^2 - 2kxy \cos \theta$ $\leftarrow (1)$

$\angle BXC = \theta$ (vertically opposite $\angle$ s equal)

Use Cosine Rule
(1)

$BC^2 = (ky)^2 + x^2 - 2 \cdot ky \cdot x \cdot \cos \theta$

$= k^2 y^2 + x^2 - 2kxy \cos \theta$ $\leftarrow (2)$

Subtracting, (1) - (2)

$AD^2 - BC^2 = k^2 x^2 + y^2 - k^2 y^2 - x^2$

Subtracting
(1)

$$\begin{aligned}
 &= (k^2-1)x^2 + (1-k^2)y^2 \\
 &= (k^2-1)x^2 - (k^2-1)y^2 \quad \left. \vphantom{\begin{aligned} &= (k^2-1)x^2 + (1-k^2)y^2 \\ &= (k^2-1)x^2 - (k^2-1)y^2 \end{aligned}} \right\} (i) \\
 &= (k^2-1)(x^2-y^2)
 \end{aligned}$$

$$\begin{aligned}
 \text{iv) } DB^2 &= (kx+x)^2 \\
 &= (x(k+1))^2 \\
 &= x^2(k+1)^2
 \end{aligned}$$

$$\begin{aligned}
 AC^2 &= (y+ky)^2 \\
 &= (y(1+k))^2 \\
 &= y^2(k+1)^2
 \end{aligned}$$

Find DB^2 or AC^2
(1)

$$\begin{aligned}
 \therefore DB^2 - AC^2 &= (k+1)^2 x^2 - (k+1)^2 y^2 \\
 &= (k+1)^2 (x^2 - y^2)
 \end{aligned}$$

$$\begin{aligned}
 \text{Ratio required} &= \frac{(k^2-1)\cancel{(x^2-y^2)}}{(k+1)^2\cancel{(x^2-y^2)}} \\
 &= \frac{(k-1)(k+1)}{(k+1)^2} \\
 &= \frac{k-1}{k+1}
 \end{aligned}$$

Answer - fully
simplified
(1)